

space	probe	per-ev.	$f^*(2\sigma)$, median over draws			SparKer range
			Maha.	MMD	SparKer	
SupCon	0.930	0.01	0.057	0.020	0.018	0.015–0.021
→ res (residual)	0.923	0.23	0.023	0.033	0.022	0.012–0.039
→ res (concat)	0.957	0.12	0.024	0.018	0.019	0.016–0.021
→ res-nplm (residual)	0.926	0.22	0.023	0.029	0.017	0.015–0.019
→ res-nplm (concat)	0.946	0.10	0.025	0.020	0.019	0.016–0.021
SupCon+SIGReg ($\lambda=5$)	0.912	0.02	$> 0.1^4$	0.031	0.037	0.032–0.042
→ res (residual)	0.902	0.01	$> 0.1^3$	0.044	0.069	0.044–0.091
→ res (concat)	0.949	0.01	$> 0.1^3$	0.035	0.042	0.027–0.060
→ res-nplm (residual)	0.921	0.16	0.044	0.038	0.022	0.017–0.025
→ res-nplm (concat)	0.943	0.06	$> 0.1^2$	0.033	0.030	0.020–0.036
NPLM-dist. (sup.)	0.788	0.17	0.038	0.039	0.027	0.024–0.040
→ res (residual)	0.884	0.03	$> 0.1^2$	0.046	0.046	0.044–0.077
→ res (concat)	0.908	0.03	$> 0.1^2$	0.042	0.043	0.033–0.059
→ res-nplm (residual)	0.841	0.01	$> 0.1^2$	0.044	0.039	0.024–0.049
→ res-nplm (concat)	0.879	0.09	0.047 ¹	0.045	0.031	0.018–0.043
NPLM-dist.+classwise SIGReg	0.938	0.21	0.032	0.033	0.027	0.024–0.029
→ res (residual)	0.917	0.03	$> 0.1^2$	0.037	0.048	0.039–0.084
→ res (concat)	0.958	0.03	0.034 ¹	0.035	0.047	0.037–0.078
→ res-nplm (residual)	0.948	0.07	0.034	0.036	0.043	0.028–0.079
→ res-nplm (concat)	0.966	0.17	0.023	0.035	0.037	0.018–0.046

Table 1: Pre-discovery: minimum injected fraction for a 2σ dataset-level detection, with and without the residual construction. f^* is the smallest fraction of the held-out class injected into a 5000-point corpus at which the median expected significance of the test reaches 2σ ; lower is better. The probe (oracle) and per-event power are shown for reference; the last column is SparKer’s range across draws. *Coverage*: CIFAR-10, leakage-free lineage, single holdout; draws 4, 7, 8, 9; 20/20 spaces on every draw; 200 null / 50 signal toys per fraction, $N_D=5000$; median over draws, superscript = draws where the test never reached 2σ by $f=0.1$.

Sigma-currency tables (exps 146/148), CIFAR-10 leakage-free lineage — generated September 1, 2026

$f^*(2\sigma)$ = smallest injected fraction at which the median expected significance of a dataset-level test reaches 2σ (200 null / 50 signal toys, $N_D=5000$); lower is better. Post tables: bold = best test per arm; † = the derived cut declined at some fraction below the crossing (the space is the frozen one there); purity ‘–’ = declined at $f=0.03$; ‘(n/6)’ = arm still running.

arm	draw	frozen SparKer	dist/95th	dist/derived	np/derived
SupCon	h4	0.015	0.027 (MMD)	0.014 (MMD) [†]	0.014 (MMD) [†]
	h7	0.021	0.035 (SparKer (anch.))	0.033 (SparKer (anch.))	0.018 (SparKer)
	h8	0.018	0.016 (Eucl. (anch.))	0.017 (MMD)	0.011 (Eucl. (anch.))
	h9	0.018	0.008 (Eucl.)	0.011 (Eucl. (anch.))	0.009 (SparKer)
SupCon+SIGReg ($\lambda=5$)	h4	0.042	0.021 (Eucl. (anch.))	0.028 (MMD)	0.028 (Eucl.)
	h7	0.034	0.021 (Eucl. (anch.))	0.036 (SparKer) [†]	0.036 (SparKer) [†]
	h8	0.040	0.015 (MMD)	0.021 (MMD)	0.020 (Eucl. (anch.))
	h9	0.032	0.016 (Eucl. (anch.))	0.026 (MMD) [†]	0.026 (MMD) [†]
NPLM-dist. (sup.)	h4	0.040	0.024 (Eucl. (anch.))	0.037 (Eucl. (anch.))	0.033 (Eucl. (anch.))
	h7	0.029	0.022 (Eucl. (anch.))	0.039 (Eucl. (anch.))	0.021 (Eucl. (anch.))
	h8	0.024	0.016 (SparKer (anch.))	0.015 (Eucl. (anch.))	0.016 (Eucl. (anch.))
	h9	0.024	0.009 (SparKer (anch.))	0.020 (Eucl. (anch.))	0.020 (Eucl. (anch.))
NPLM-dist.+classwise SIGReg	h4	0.025	0.019 (Eucl. (anch.))	0.020 (Eucl. (anch.))	0.020 (Eucl. (anch.))
	h7	0.029	0.018 (Eucl. (anch.))	0.027 (SparKer (anch.))	0.011 (Eucl. (anch.))
	h8	0.024	0.015 (Eucl. (anch.))	0.012 (Eucl. (anch.))	0.009 (Eucl. (anch.))
	h9	0.028	0.016 (Maha.)	0.020 (Eucl. (anch.))	0.020 (Eucl. (anch.))

Table 2: Does the discovery loop beat the frozen space? Frozen-space SparKer $f^*(2\sigma)$ against the best post-discovery test (named in parentheses) under three loop variants: the campaign’s distance pool with the 95th-percentile cut, the distance pool with the derived cut, and the density-ratio pool with the derived cut (the paper’s construction). Bold = the loop wins on that draw. *Coverage:* CIFAR-10, four parents x draws 4, 7, 8, 9; 48 complete (arm, draw, variant) cells, the loop beats the frozen space in 38; [†] = the derived cut had declined at the crossing fraction, so that number is the frozen space’s and is not counted as a win; ‘-’ = variant not run or still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	0.00	0.063	> 0.1	0.072	0.027	0.036	0.029
→ res (residual)	0.039	0.04	0.074	0.026	0.039	0.043	0.047	0.026
→ res-nplm (residual)	0.016	0.08	0.082	0.029	0.065	0.045	0.037	0.031
SupCon+SIGReg ($\lambda=5$)	0.042	0.01	0.033	0.021	> 0.1	0.027	0.031	0.039
→ res (residual)	0.091	0.01	0.068	0.043	0.035	0.036	0.044	0.045
→ res-nplm (residual)	0.023	0.10	0.027	0.017	0.077	0.028	0.029	0.037
NPLM-dist. (sup.)	0.040	0.10	> 0.1	0.024	0.087	0.046	0.046	0.043
→ res (residual)	0.077	0.01	> 0.1	> 0.1	0.081	0.035	0.043	0.041
→ res-nplm (residual)	0.049	0.01	> 0.1	0.047	0.087	0.062	0.083	0.044
NPLM-dist.+classwise SIGReg	0.025	0.20	0.034	0.019	0.033	0.044	0.028	0.024
→ res (residual)	0.084	0.01	0.027	0.027	0.033	0.033	0.029	0.027
→ res-nplm (residual)	0.079	0.03	0.043	0.022	0.037	0.036	0.047	0.041

Table 3: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, distance pool, 95th-pct. cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	–	> 0.1	–	> 0.1	0.014 †	0.017†	–
→ res (residual)	0.039	0.03	0.042	0.039	0.068	0.039	0.040	0.032
→ res-nplm (residual)	0.016	0.01	0.047	0.039	0.049	0.034	0.027	0.024
SupCon+SIGReg ($\lambda=5$)	0.042	0.00	0.038	0.062	> 0.1	0.028	0.047	0.044
→ res (residual)	0.091	0.00	> 0.1	> 0.1	0.078	0.038	0.049	0.087
→ res-nplm (residual)	0.023	0.07	0.072	0.024	> 0.1	0.039	0.040	0.026
NPLM-dist. (sup.)	0.040	–	0.077	0.037	0.072	0.045	0.043	0.043
→ res (residual)	0.077	0.02	> 0.1	> 0.1	> 0.1	0.038	0.052	0.060
→ res-nplm (residual)	0.049	–	0.089	0.065	0.089	0.080	0.049 †	0.074
NPLM-dist.+classwise SIGReg	0.025	–	0.028†	0.020	0.074	0.029†	0.028†	0.021
→ res (residual)	0.084	–	> 0.1	0.050	0.083	0.041	0.047	0.050
→ res-nplm (residual)	0.079	–	0.049	0.050	0.046	0.043	0.067	0.050

Table 4: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, distance pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	–	> 0.1	–	> 0.1	0.014 [†]	0.017 [†]	–
→ res (residual)	0.039	0.21	0.033	0.015	0.024	0.041	0.033	0.029
→ res-nplm (residual)	0.016	0.14	0.035	0.020	0.039	0.041	0.037	0.034
SupCon+SIGReg ($\lambda=5$)	0.042	0.13	0.028	0.030	0.083	0.034	0.042	0.030
→ res (residual)	0.091	0.15	0.082	0.030	0.033	0.042	0.060	0.047
→ res-nplm (residual)	0.023	0.12	0.044	0.018	0.074	0.037	0.041	0.021
NPLM-dist. (sup.)	0.040	–	0.087	0.033	0.084	0.049	0.073	0.042
→ res (residual)	0.077	0.43	0.056	0.020	0.029	0.046	0.046	0.034
→ res-nplm (residual)	0.049	–	0.089	0.046	0.089	0.080	0.049 [†]	0.062
NPLM-dist.+classwise SIGReg	0.025	–	0.028 [†]	0.020	0.072	0.029 [†]	0.027 [†]	0.036
→ res (residual)	0.084	–	0.079	0.050	0.046	0.044	0.045	0.050
→ res-nplm (residual)	0.079	–	0.044	0.050	0.046	0.043	0.057	0.050

Table 5: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-10 holdout 4, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.00	> 0.1	> 0.1	0.043	0.047	0.052	0.035
SupCon+SIGReg ($\lambda=5$)	0.034	0.03	0.042	0.021	> 0.1	0.056	0.040	0.043
NPLM-dist. (sup.)	0.029	0.08	0.082	0.022	0.038	0.044	0.029	0.023
NPLM-dist.+classwise SIGReg	0.029	0.17	0.029	0.018	0.024	0.037	0.028	0.025

Table 6: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.00	0.071	> 0.1	0.036	0.056	0.040	0.033
SupCon+SIGReg ($\lambda=5$)	0.034	–	0.079	–	0.090	0.079	0.036 †	–
NPLM-dist. (sup.)	0.029	0.05	> 0.1	0.039	0.044	0.042	0.042	0.041
NPLM-dist.+classwise SIGReg	0.029	0.10	0.042	0.029	0.035	0.052	0.040	0.027

Table 7: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.68	0.022	0.020	0.021	0.052	0.018	0.020
SupCon+SIGReg ($\lambda=5$)	0.034	–	0.079	–	0.090	0.072	0.036 †	–
NPLM-dist. (sup.)	0.029	0.18	0.044	0.021	0.033	0.043	0.041	0.029
NPLM-dist.+classwise SIGReg	0.029	0.67	0.043	0.011	0.018	0.049	0.033	0.019

Table 8: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, density-ratio pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.039	0.016	0.023	0.018	0.021	0.017
SupCon+SIGReg ($\lambda=5$)	0.040	0.01	0.026	0.019	> 0.1	0.015	0.026	0.023
NPLM-dist. (sup.)	0.024	0.16	0.037	0.017	0.019	0.019	0.021	0.016
NPLM-dist.+classwise SIGReg	0.024	0.18	0.017	0.015	0.022	0.022	0.017	0.015

Table 9: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.052	> 0.1	0.038	0.017	0.031	0.027
SupCon+SIGReg ($\lambda=5$)	0.040	0.01	0.024	0.076	0.062	0.021	0.029	0.029
NPLM-dist. (sup.)	0.024	0.59	0.063	0.015	0.025	0.023	0.023	0.017
NPLM-dist.+classwise SIGReg	0.024	0.55	0.042	0.012	0.039	0.036	0.035	0.021

Table 10: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.45	0.014	0.011	0.015	0.019	0.017	0.014
SupCon+SIGReg ($\lambda=5$)	0.040	0.22	0.024	0.020	0.024	0.025	0.032	0.035
NPLM-dist. (sup.)	0.024	0.64	0.046	0.016	0.041	0.024	0.018	0.017
NPLM-dist.+classwise SIGReg	0.024	0.73	0.075	0.009	0.037	0.044	0.040	0.021

Table 11: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, density-ratio pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.02	0.008	0.010	0.017	0.018	0.010	0.009
SupCon+SIGReg ($\lambda=5$)	0.032	0.01	0.025	0.016	0.040	0.040	0.025	0.025
NPLM-dist. (sup.)	0.024	0.14	0.017	0.014	0.015	0.018	0.014	0.009
NPLM-dist.+classwise SIGReg	0.028	0.17	0.016	0.022	0.016	0.029	0.017	0.017

Table 12: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.013	0.011	0.025	0.020	0.014	0.018
SupCon+SIGReg ($\lambda=5$)	0.032	–	0.047	> 0.1	0.048	0.026 †	0.027†	0.050
NPLM-dist. (sup.)	0.024	0.82	0.036	0.020	0.024	0.029	0.021	0.027
NPLM-dist.+classwise SIGReg	0.028	0.51	0.026	0.020	0.028	0.036	0.026	0.020

Table 13: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.28	0.010	0.013	0.016	0.025	0.009	0.010
SupCon+SIGReg ($\lambda=5$)	0.032	–	0.047	0.050	0.045	0.026 †	0.027†	0.050
NPLM-dist. (sup.)	0.024	0.84	0.027	0.020	0.023	0.029	0.023	0.020
NPLM-dist.+classwise SIGReg	0.028	0.86	0.044	0.020	0.025	0.038	0.025	0.020

Table 14: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, density-ratio pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.